Day 7 Of Communication System Class

2 Amplitude Modulation
2.1 Amplitude Modulation
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2.3 Different Types OF AM Transmission
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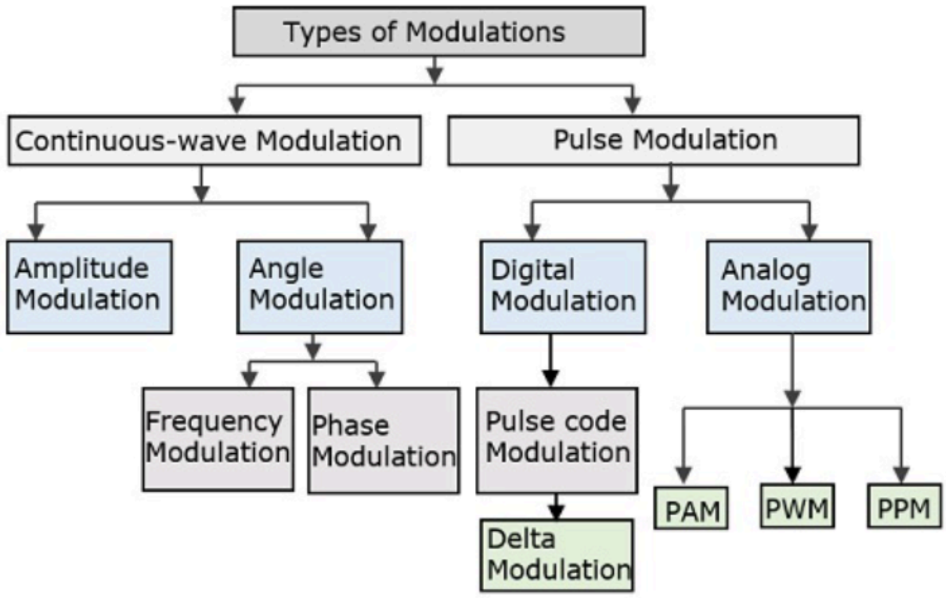

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2.1 Amplitude Modulation

AMPLITUDE MODULATION :

In amplitude modulation amplitude of a carrier signal is varied by the modulating signal, while other parameter will remain constant.

Equation of AM signal :

Instantaneous value of modulating signal

$$e_m = E_m \sin \omega_m t$$

and instantaneous value of carrier signal

$$e_c = E_c \sin \omega_c t$$

where E_m and E_c are maximum amplitude.

Create a new mathematical expression for complete modulated wave

$$E_{AM} = E_c + e_m = E_c + E_m \sin \omega_m t$$

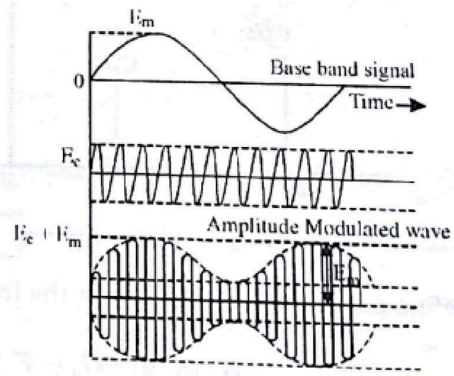

Fig. 3.7

Modulation and Noise

Therefore, instantaneous value of amplitude modulated wave can be given as

$$e_{AM} = E_{AM} \sin \theta$$
$$= E_{AM} \sin \omega_c t$$
$$= (E_c + E_m \sin \omega_m t) \sin \omega_c t$$

Modulation index

Modulation index is given by m, where

$$m = \frac{E_m}{E_c}$$ (modulation index) or modulation factor or degree of modulation.

m is a number lying between 0 and 1

2.2 Representation OF AM Wave

Representation of AM Wave :

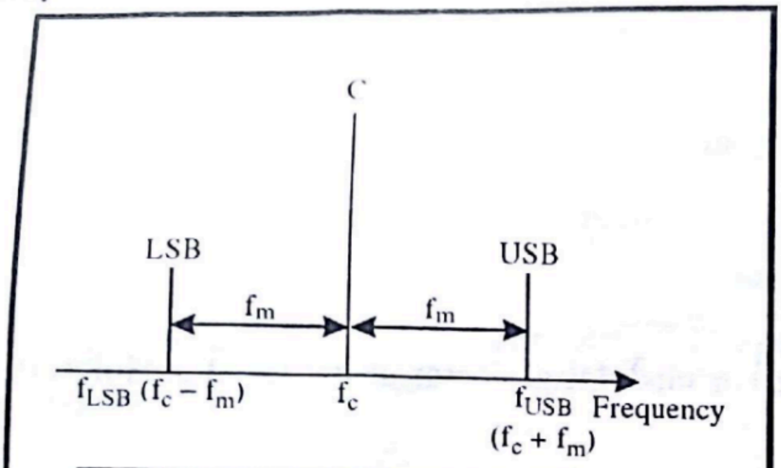

Fig. 3.8. Frequency spectrum of AM wave

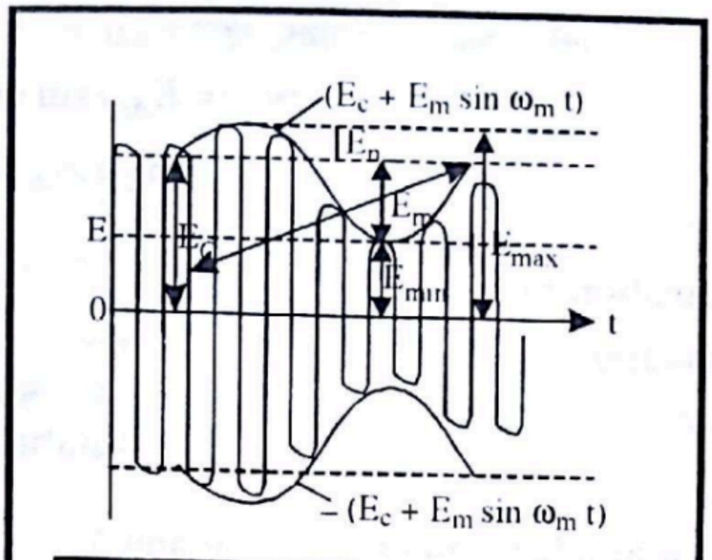

Fig. 3.9. Amplitude Modulated wave

Problem 2. *An audio frequency signal* $10 \sin 2\pi \times 500t$ *is used to amplitude modulated carrier of* $50 \sin 2\pi \times 10^5 t$. *Calculate*

(i) Modulation index

(ii) Side band frequency

(iii) Amplitude of each side band

(iv) Bandwidth requirement

(v) Total power delivered if $R = 600\ \Omega$.

Solution : We have $e_m = 10 \sin 2\pi \times 500t$

$$e_c = 50 \sin 2\pi \times 10^5 t$$

(i) Modulation index $= \dfrac{E_m}{E_c} = \dfrac{10}{50} = \dfrac{1}{5}$

(ii) Side band frequency $f_c = 10^5$ Hz, $f_m = 500$ Hz

So,
$$f_{USB} = (10^5 + 500) = 1.005 \times 10^5 \text{ Hz}$$
$$F_{LSB} = (10^5 - 500) = 9.5 \times 10^3 \text{ Hz.}$$

(iii) Amplitude of each side band $= \dfrac{mE_c}{2} = \dfrac{0.2 \times 50}{2} = 5$ volt.

(iv) Bandwidth requirement $= 2f_m = 2 \times 500 = 1000$ Hz.

(v) Total power delievered if $R = 600\ \Omega$.

$$P_T = P_c\left(1 + \frac{m^2}{2}\right)$$

So,
$$P_T = \frac{E_c^2}{2R}\left(1 + \frac{m^2}{2}\right) = \frac{(50)^2}{2 \times 600}\left(\frac{(0.2)^2}{2}\right) = 2.175 \text{ watt.}$$

2.3 Different Types OF AM Transmission

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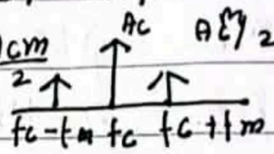
AM Transmission

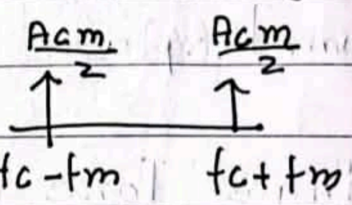
1) Full Carrier (FC) w/ double Side band



* Power मा Pay

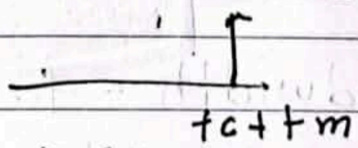
(FCDSB)

2) Supress Carrier w/ Double Side Band



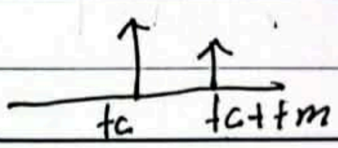
(SC-DSB)

3) Suppress Carrier w/ Single Side band



(SC-SSB) → SC SSB USB or LSB

4) Full Carrier w/ Single Side Band

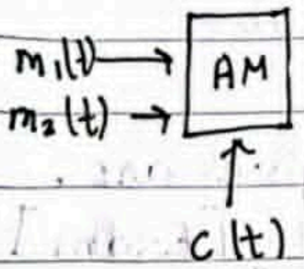


(FC-SSB)

2.4 Double Tone AM

[Exam] Double Tone AM

1) 2M1C



$$m_1(t) = A_{m_1} \sin(\omega_{m_1} t)$$

$$m_2(t) = A_{m_2} \sin(\omega_{m_2} t)$$

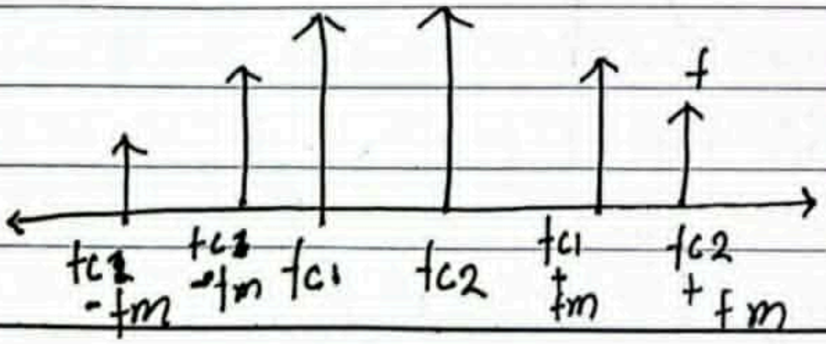
$$c(t) = A_c \sin(\omega_c t)$$

$$\Rightarrow S_{AM} = \sin(\omega_c t)\left[A_c + m_1 \cancel{+ m_2}\right] + \sin(\omega_c t)\left[A_c + m_2\right]$$

$$\sin\cancel{(\omega_c t)\left[A_c + A_{mr}\right.}$$

2) 1M2C

$$S_{AM} = \sin(\omega_{c1} t)\left[A_{c1} + m\right] + \sin(\omega_{c2} t)\left[A_{c2} + m\right]$$



[Most Probable Question]

Q. How to generate AM signal with circuit. [Exam]

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Q. AM is very noisy. give ans.
- A लाई प्लाट गर [Exam]
- Swing of A $[A_c + m(t)] \sin(\omega_c t)$

Q. Different types of AM generators. modulators and demodulators.